Embedded Control for Robotics

Lecture # 05b **Time Response**

- Book: Classical and Modern Control with Microcontrollers, Ch 2, 4

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Analysis of Transfer Function

1. **Pole/Zero**
   - Why??
     - Pole/Zero Stability
2. **Time dynamic analysis**
     - DC Gain via step response
     - Time constant via step response
3. **Frequency Analysis**
     - Phase Shift
     - Amplitudes

•$G(s) = G(j\omega)$, ω is the frequency

The absolute value of s, $|s| = \sqrt{a^2 + b^2}$

•$|s|$ is often called ω_0

Characteristic Equation

set the denominator of the transfer function to zero to obtain the characteristic equation

$$G(s) = \frac{Y(s)}{U(s)} = \frac{b_m s^m + b_{m-1} s^{m-1} + \cdots + b_1 s + b_0}{s^n + a_{n-1} s^{n-1} + \cdots + a_1 s + a_0}$$

$$s^n + a_{n-1} s^{n-1} + \ + a_1 s + a_0 = 0$$

It n roots which determines the stability of the system

Example # 1 Poles/Zeros

- Figure out the TF of $\frac{d^2y}{dt^2} + 3\frac{dy}{dt} + 2y = u + \frac{du}{dt}$

Taking the Laplace transform of this equation, ignoring terms due to initial conditions, we obtain

$$s^2Y(s) + 3sY(s) + 2Y(s) = U(s) + sU(s)$$

This equation can be written as

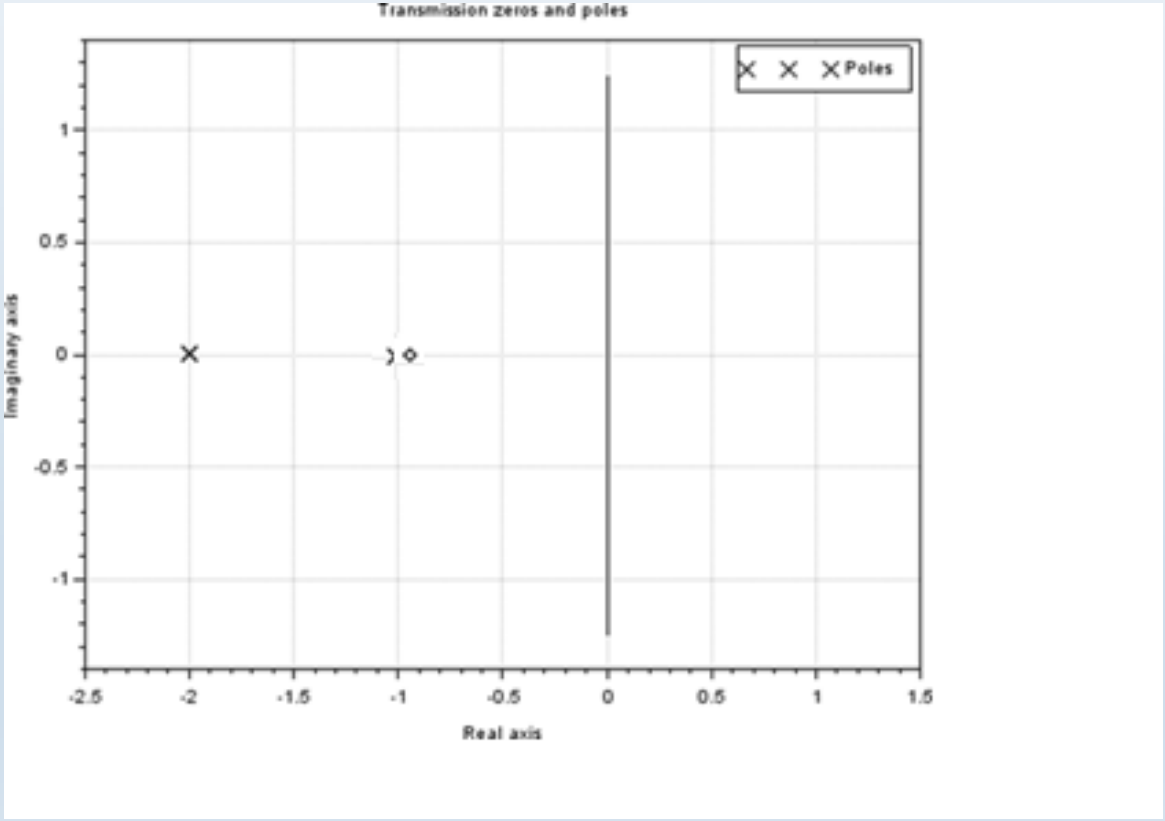
$$Y(s) = \left[\frac{s+1}{s^2+3s+2}\right]U(s)$$

The transfer function of this system is therefore given by

$$P(s) = \frac{s+1}{s^2+3s+2}$$

Zero -1

Poles -2,-1



Stability …?

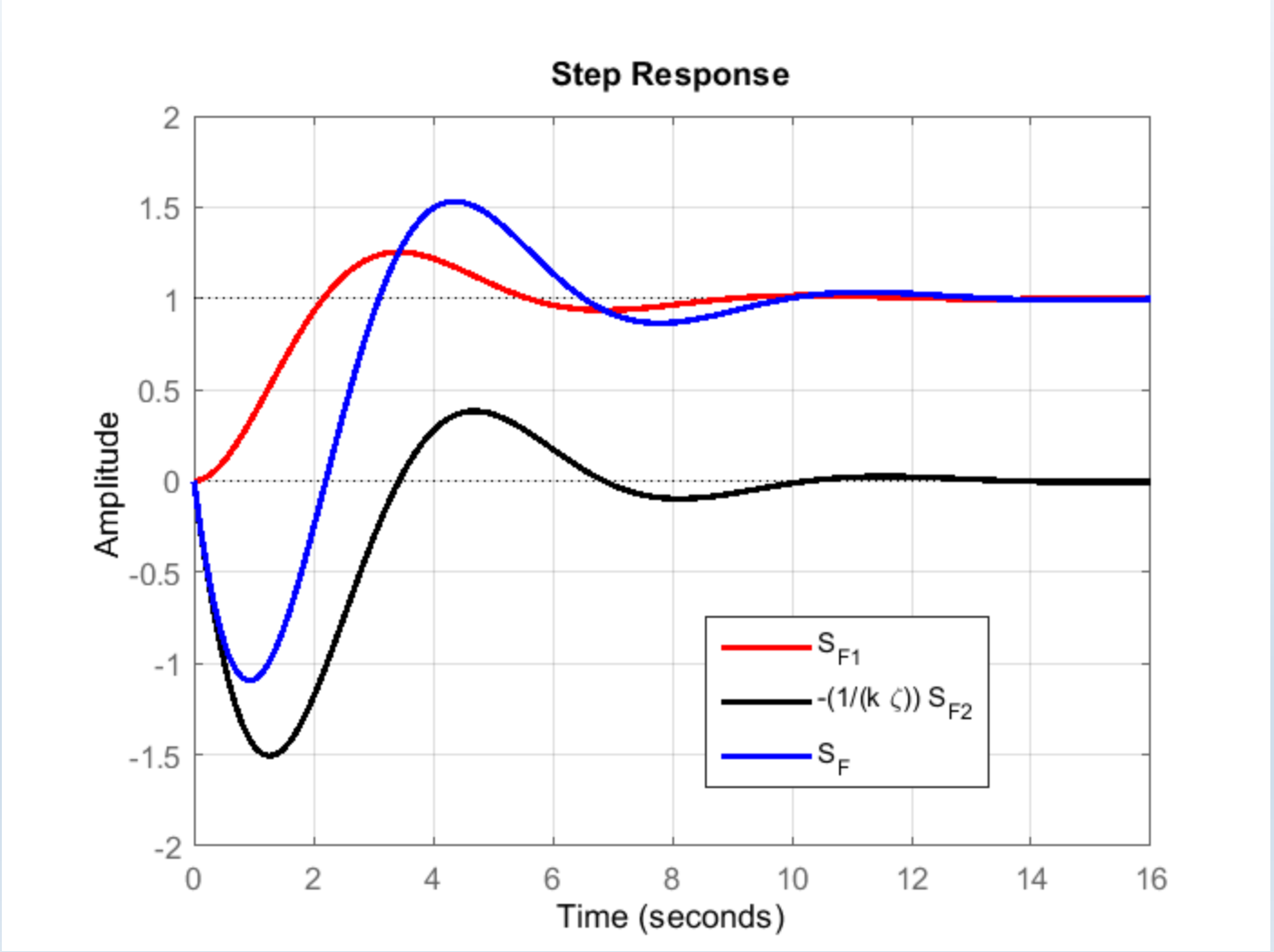
- Stability of the following transfer functions:

$$(a) \quad P(s) = \frac{(s-1)}{(s+2)(s^2+4)}$$

$$(b) \quad P(s) = \frac{(s-1)}{(s+2)(s+4)}$$

- Unstable: a : Two real poles at the same locations
- Stable: b
- Due to Positive Zeros, the system initially moves in the opposite direction of its steady-state response.

Positive Zero Response



Python Code

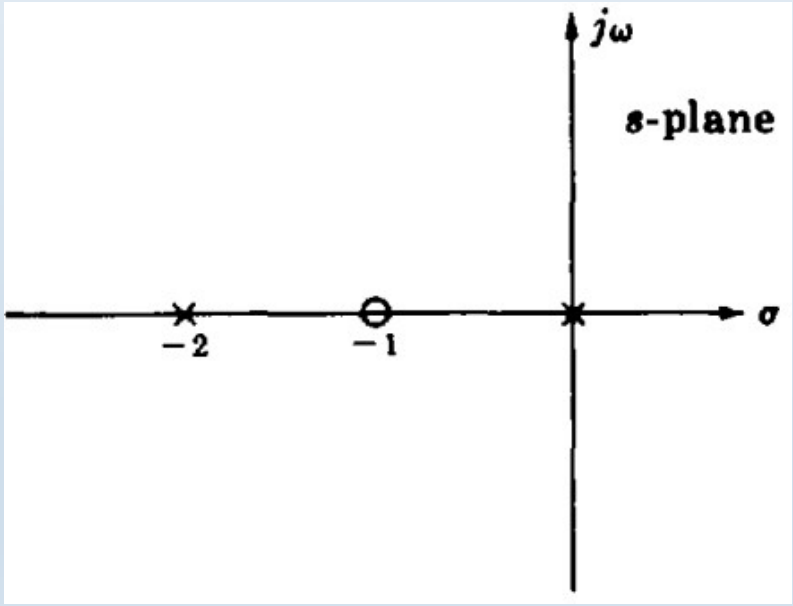
- **Transfer function.**
- **Plot the Pole-zero plot**
- **Step Response**

$$P(s) = \frac{(s-1)}{(s+2)(s^2+4)}$$

```
import control
N = 400
Ts=0.0005
t = [n*Ts for n in range(N)]
num = [1, -1]
den = [1, 2, 4, 8]
g = control.TransferFunction(num, den)
control.pole_zero_plot(g)
response = control.step_response(g, T=t)
response.plot()
```

Plot

- What is the TF of a system with a gain factor of 2 and a pole-zero map in the s-plane as shown:



Time domain dynamic response

- In control systems, **time is used as an independent variable** to evaluate the state and output responses, called the **time response.**

- **A reference input signal is applied to a system**, and **the performance of the system** is evaluated by studying the system response in the **time domain.**



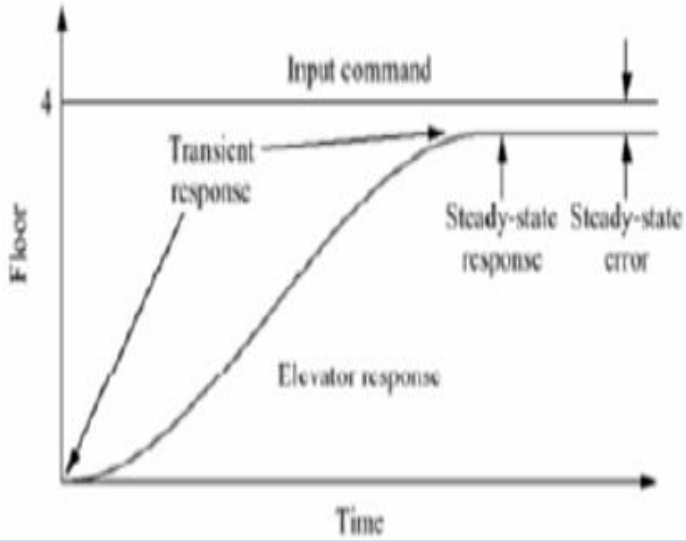
Time domain dynamic response

The time response is divided into two parts:

the **transient response**

the **steady-state response.**

$$y(t) = y_t(t) + y_{ss}(t)$$



$y_t(t)$: ***transient response*** is defined as the part of the time response that goes to zero as time becomes very large.

$$\lim_{t \to \infty} y_t(t) = 0$$

$y_{ss}(t)$:***The steady-state response*** *is simply the part of the total response that remains after the transient has died out.* ***(final accuracy of the system)***

All real, stable control systems exhibit transient phenomena to some extent before the steady state is reached

Time constant form of TF

$$G(s) = \frac{K(\tau_{z_1}s+1)(\tau_{z_2}s+1)\dots(\tau_{z_m}s+1)}{(\tau_{p_1}s+1)(\tau_{p_2}s+1)\dots(\tau_{p_n}s+1)}$$

- Key parameters:

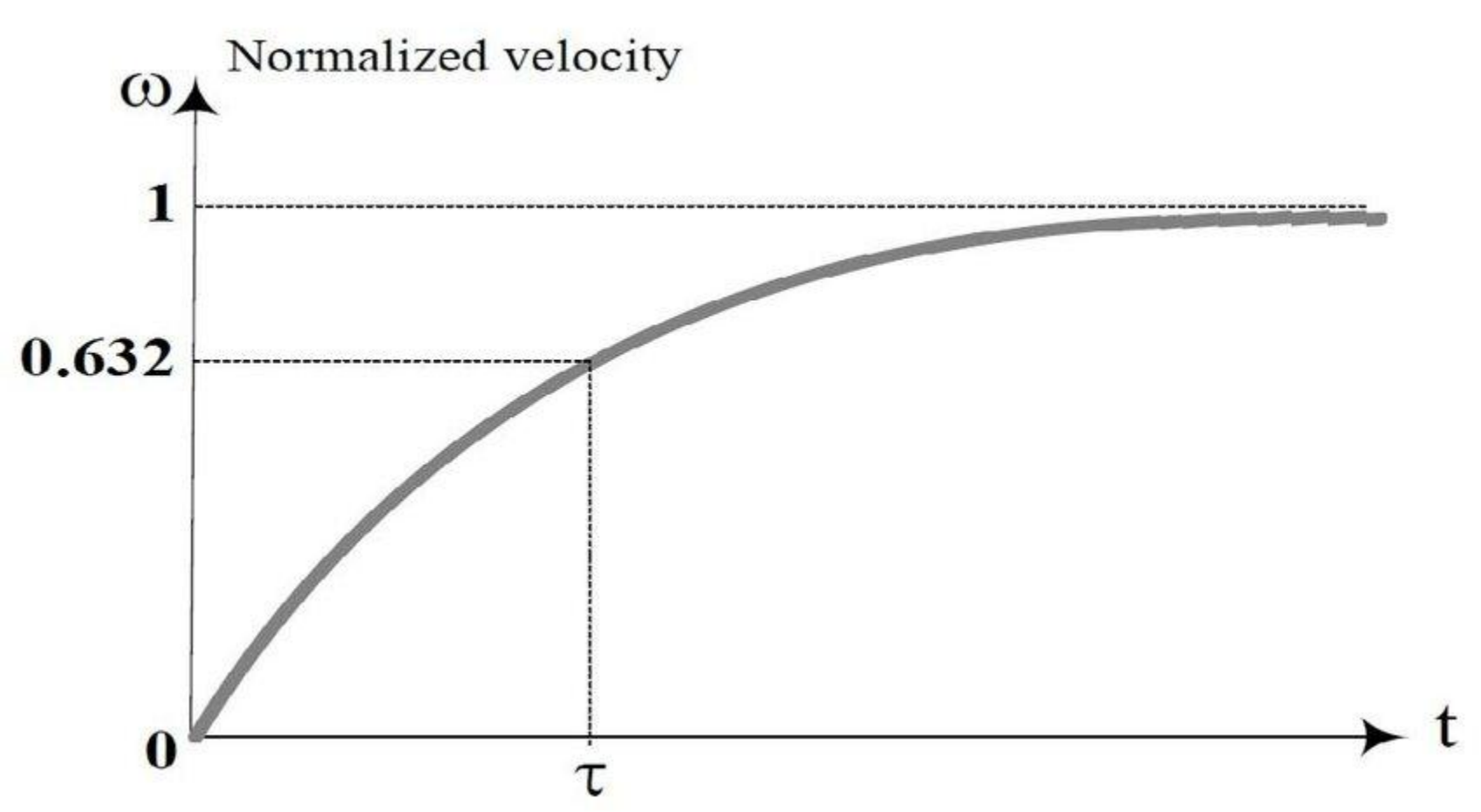
  τ = time constant

  k = DC gain

Time Constant

- Time constant (τ) represents the time it takes for the system's output to reach approximately 63.2% of its total final change after a step input.
- It's a measure of a system's "inertia" or how quickly it responds to changes;
  - a smaller time constant signifies a faster response,
  - while a larger one indicates a slower response.
-
- This value is a key transient response specification for first-order control systems.

Time Response



DC Gain

- DC gain is the ratio of the output's steady-state value to the step input's constant value.
- Representing the gain at zero frequency.
- This can be calculated from system transfer function G(s) at s = 0,
- Or by directly observing the final, constant output value in a stable system's response to a step input.
- If a system is stable, its DC gain (K) can be calculated from its step response using the following formula:

$$K_{DC} = \frac{\text{Steady-State Output}}{\text{Steady-State Input}}$$

Time Response of First-order System

Two ways of writing:

$G(s) = \frac{k}{s+a}$ Good for frequency domain,

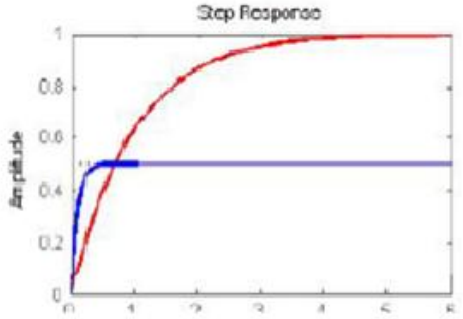
$G(s) = \frac{k}{\tau s+1}$ Good for time domain

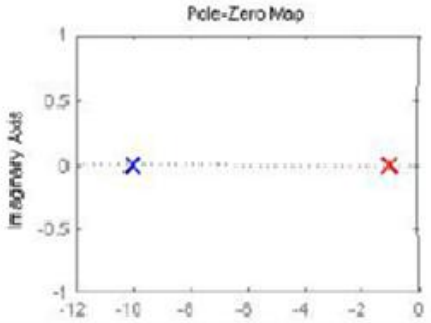
Characteristics are:

Pole

Dc-gain k

Time constant τ





Example: $G_1 = \frac{1}{s+1}$

$G_2 = \frac{5}{s+10}$

Second-order System

TF of a second-order system can be rewritten in a standard format as

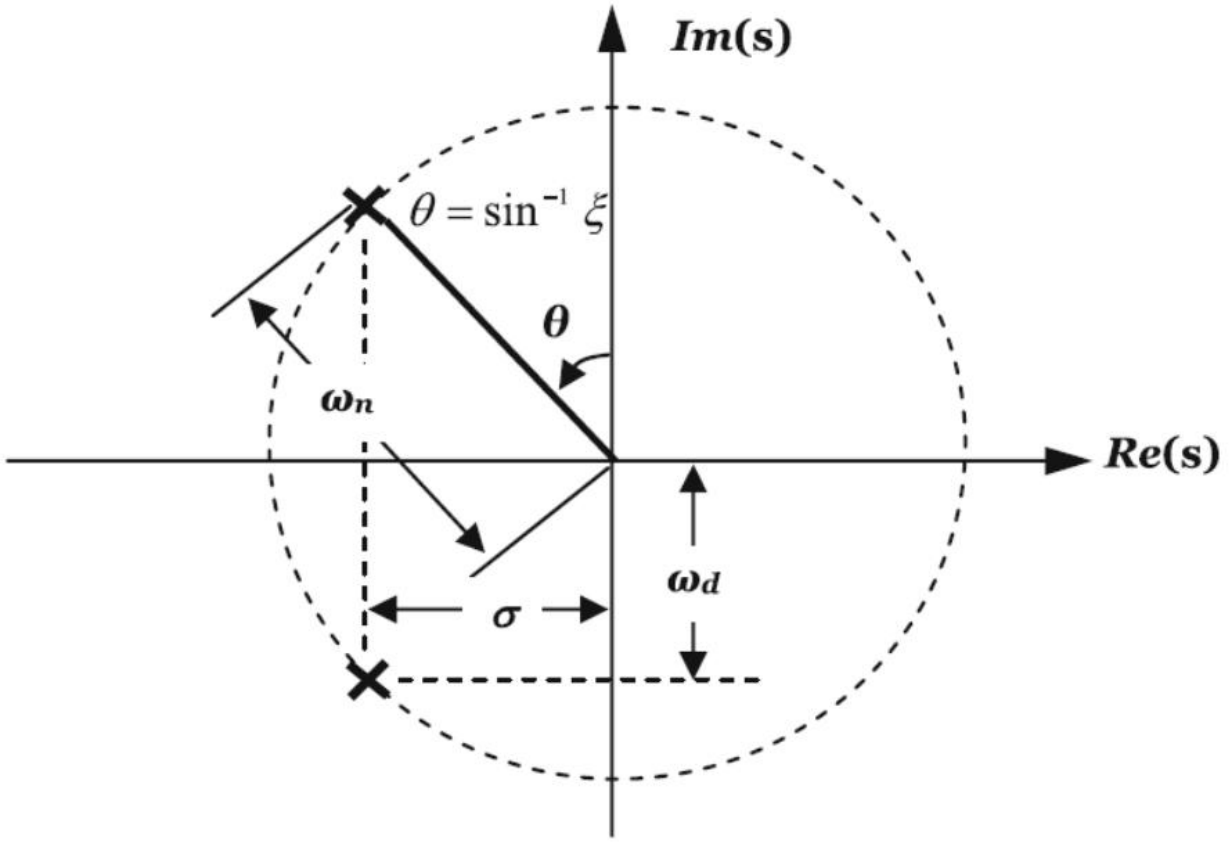
$$G(s) = K\frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$\sigma = \zeta\omega_n$$

$$\omega_d = \omega_n\sqrt{1-\zeta^2}$$

where the parameter ξ is called the **damping ratio**, ω_n is called the **undamped natural frequency**, and ω_d is actual frequency.

Relationship Between Poles and Step Response



4.4 The *s*-plane configuration for a pair of complex poles

Time response specifications of the second-order

1. **Delay time** (t_d) is the time required to reach at 50% of its final value by a time response signal during its first cycle of oscillation.

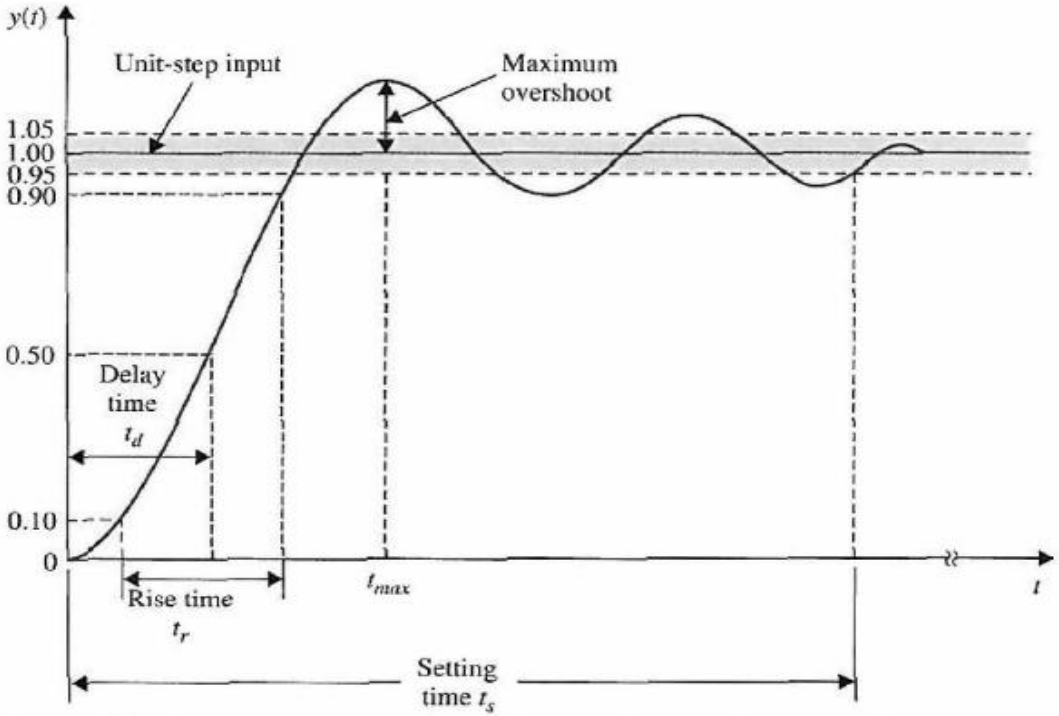


Figure 5-2 Typical unit-step response of a control system illustrating the time-domain specifications.

Time response specifications of the second-order

2. **Rise time** (t_r) is the time required to reach at final value by a under damped time response signal during its first cycle of oscillation. If the signal is over damped, then rise time is counted as the time required by the response to rise from 10% to 90% of its final value.

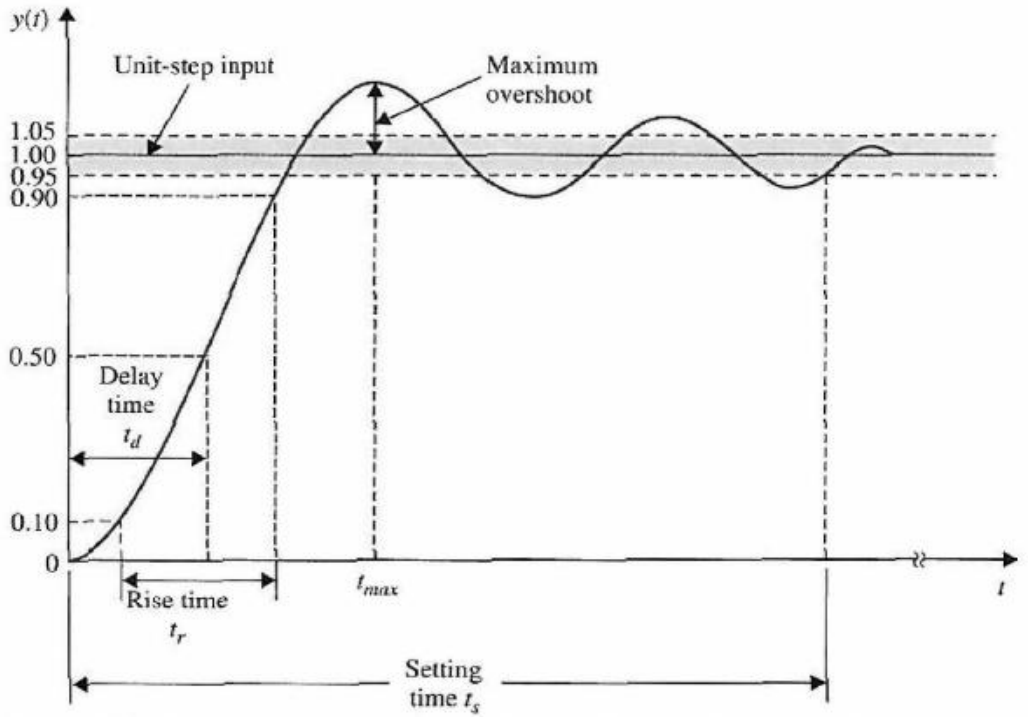


Figure 5-2 Typical unit-step response of a control system illustrating the time-domain specifications.

Time response specifications of the second-order

3. **Peak time** (t_p) is simply the time required by response to reach its first peak i.e. the peak of first cycle of oscillation, or first overshoot.

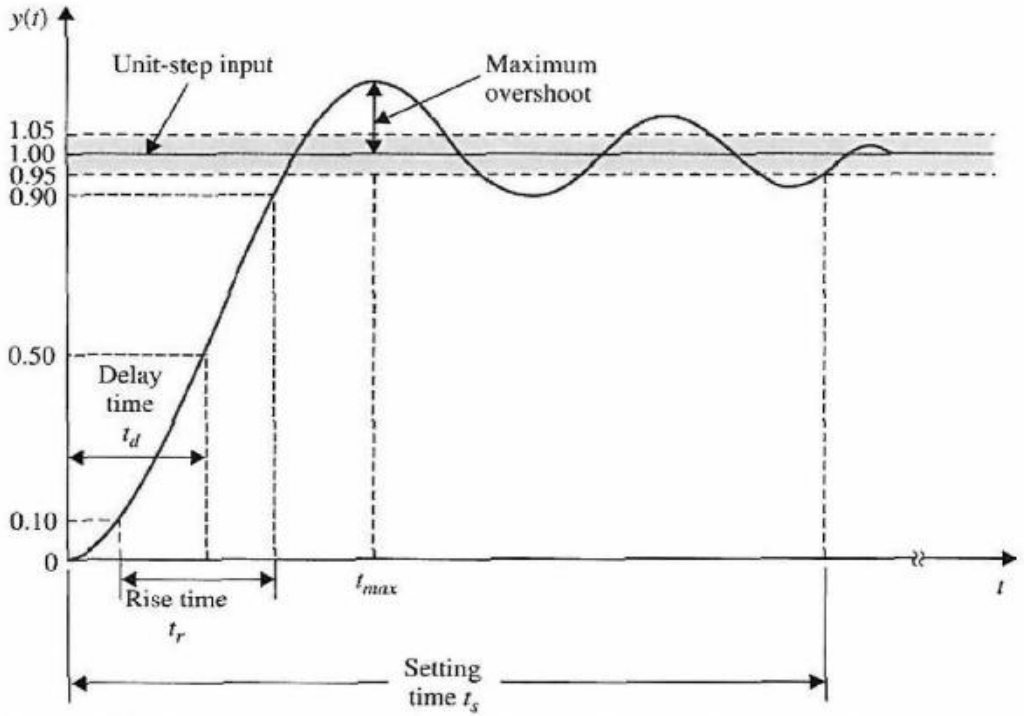


Figure 5-2 Typical unit-step response of a control system illustrating the time-domain specifications.

Time response specifications of the second-order

4. **Maximum overshoot** (M_p) is straight way difference between the magnitude of the highest peak of time response and magnitude of its steady state. Maximum overshoot is expressed in term of percentage of steady-state value of the response. As the first peak of response is normally maximum in magnitude, maximum overshoot is simply normalized difference between first peak and steady-state value of a response.

$$Maximum\ \%\ Overshoot = \frac{c(t_p) - c(\infty)}{c(\infty)} \times 100\%$$

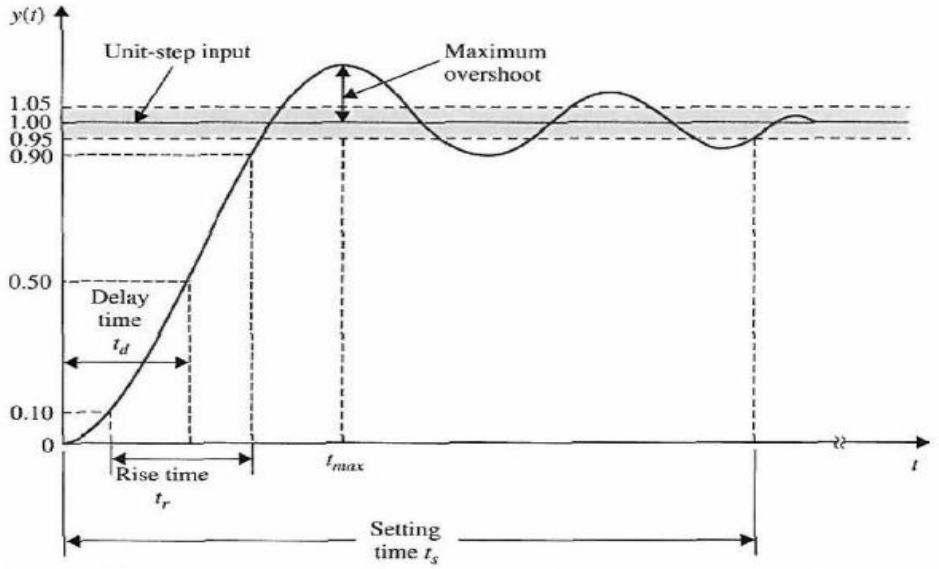


Figure 5-2 Typical unit-step response of a control system illustrating the time-domain specifications.

Time response specifications of the second-order

5. **Settling time** (t_s) is the time required for a response to become steady. It is defined as the time required by the response to reach and steady within specified range of 2% to 5% of its final value.

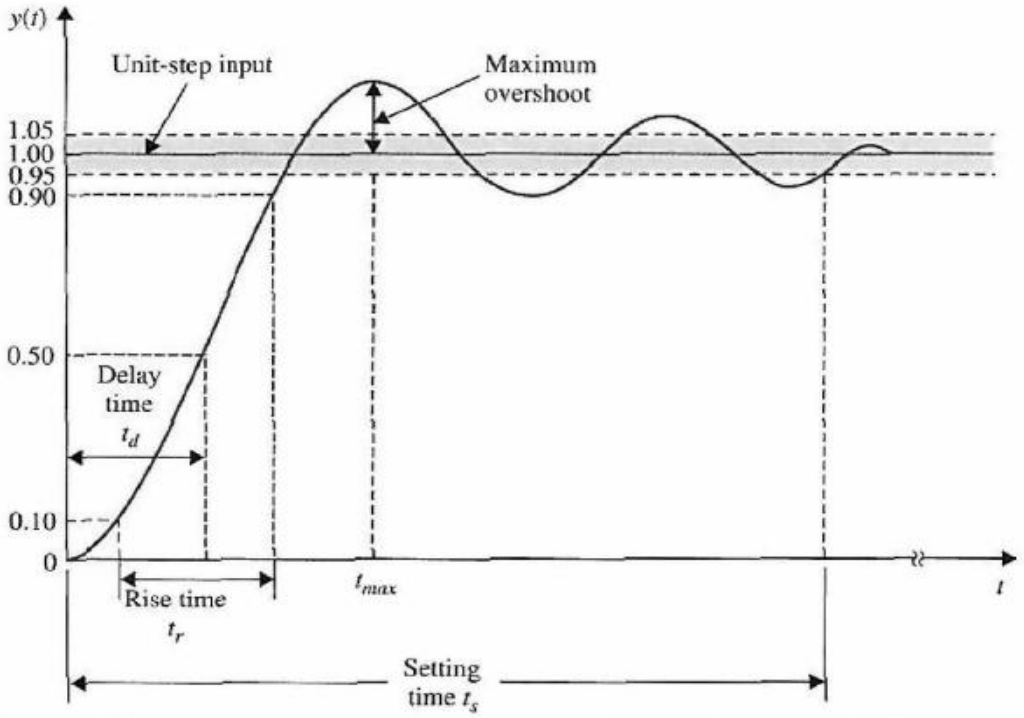


Figure 5-2 Typical unit-step response of a control system illustrating the time-domain specifications.

Time response specifications of the second-order

Steady-state error (e_{ss}) is the difference between actual output and desired output at the infinite range of time.

$$e_{ss} = \lim_{t \to \infty} [r(t) - c(t)]$$

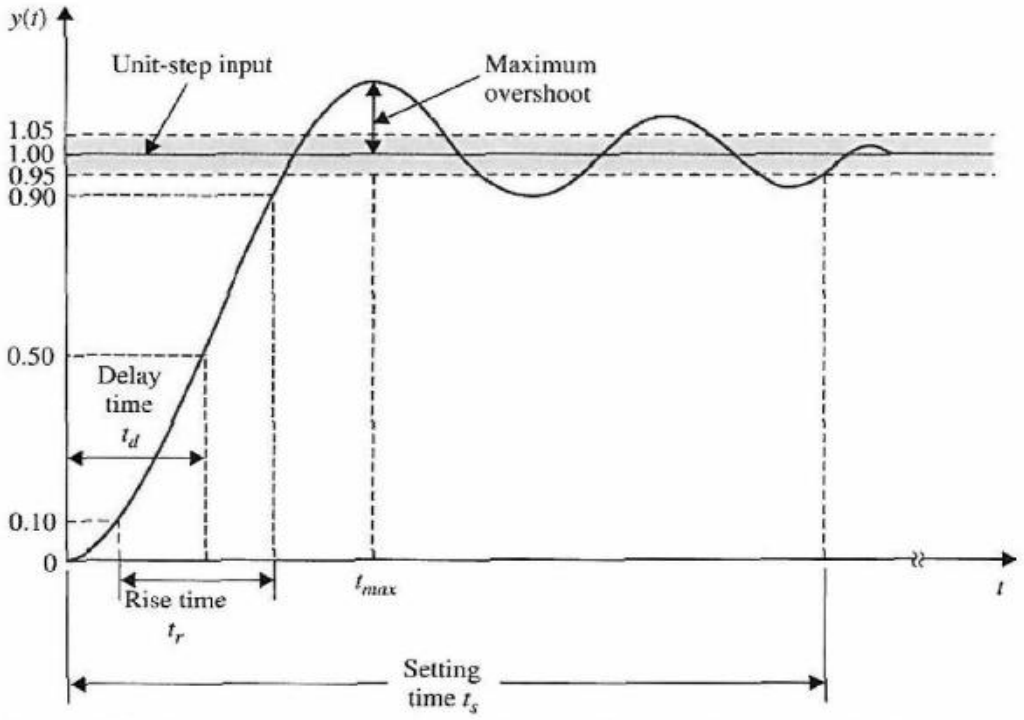


Figure 5-2 Typical unit-step response of a control system illustrating the time-domain specifications.

2nd order Eq

- Let us consider a general second order system with a transfer function of the general form:

$$H(s) = \frac{Y(s)}{X(s)} = \frac{b_2 s^2 + b_1 s + b_0}{s^2 + a_1 s + a_0}$$

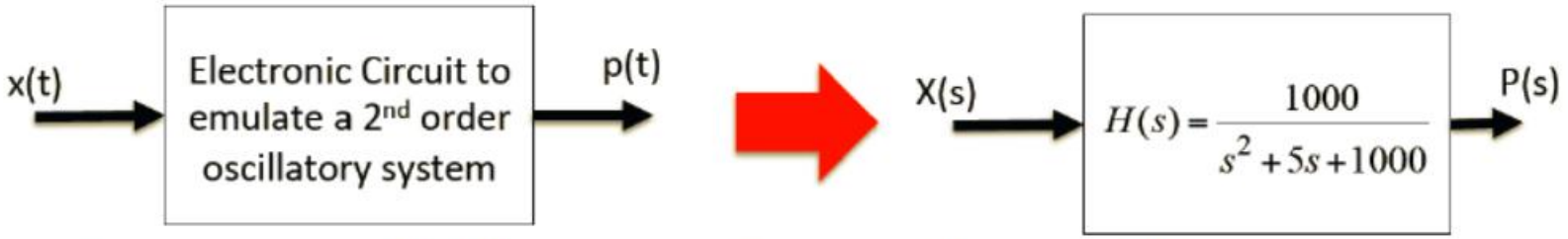
- To simplify the problem a bit, let us assuming that b2 = b1 = 0. The above equation can be rewritten as:

$$H(s) = \frac{b_0}{s^2 + a_1 s + a_0} = K \frac{{\omega_0}^2}{s^2 + 2\zeta\, \omega_0 s + {\omega_0}^2}$$

- where:
  - $\omega_0 = \sqrt{a_0}$, the resonant (or natural) frequency in rad/sec
  - $\zeta = \frac{a_1}{2\sqrt{a_0}}$, the damping factor (no unit) (pronounced as zeta)
  - $K = \frac{b_0}{a_0}$, gain of the system

Physical meaning of ω_0, ς, and K

- Let us take the transfer function H(s) of the 2nd order system used in Bulb Box as an example:



- $\omega_0 = \sqrt{a_0} = 31.62$, the resonant frequency = 5Hz
- $\zeta = \frac{a_1}{2\sqrt{a_0}} = \frac{5}{2\sqrt{1000}} = 0.079$ the damping factor (very small, ideal = 1)
- $K = \frac{b_0}{a_0} = 1$, gain of the system at DC or zero frequency

- Since the damping factor is very small (much smaller than 1), this system is highly oscillatory.

$$H(s) = \frac{b_0}{s^2 + a_1 s + a_0} = K \frac{{\omega_0}^2}{s^2 + 2\zeta\omega_0 s + {\omega_0}^2}$$

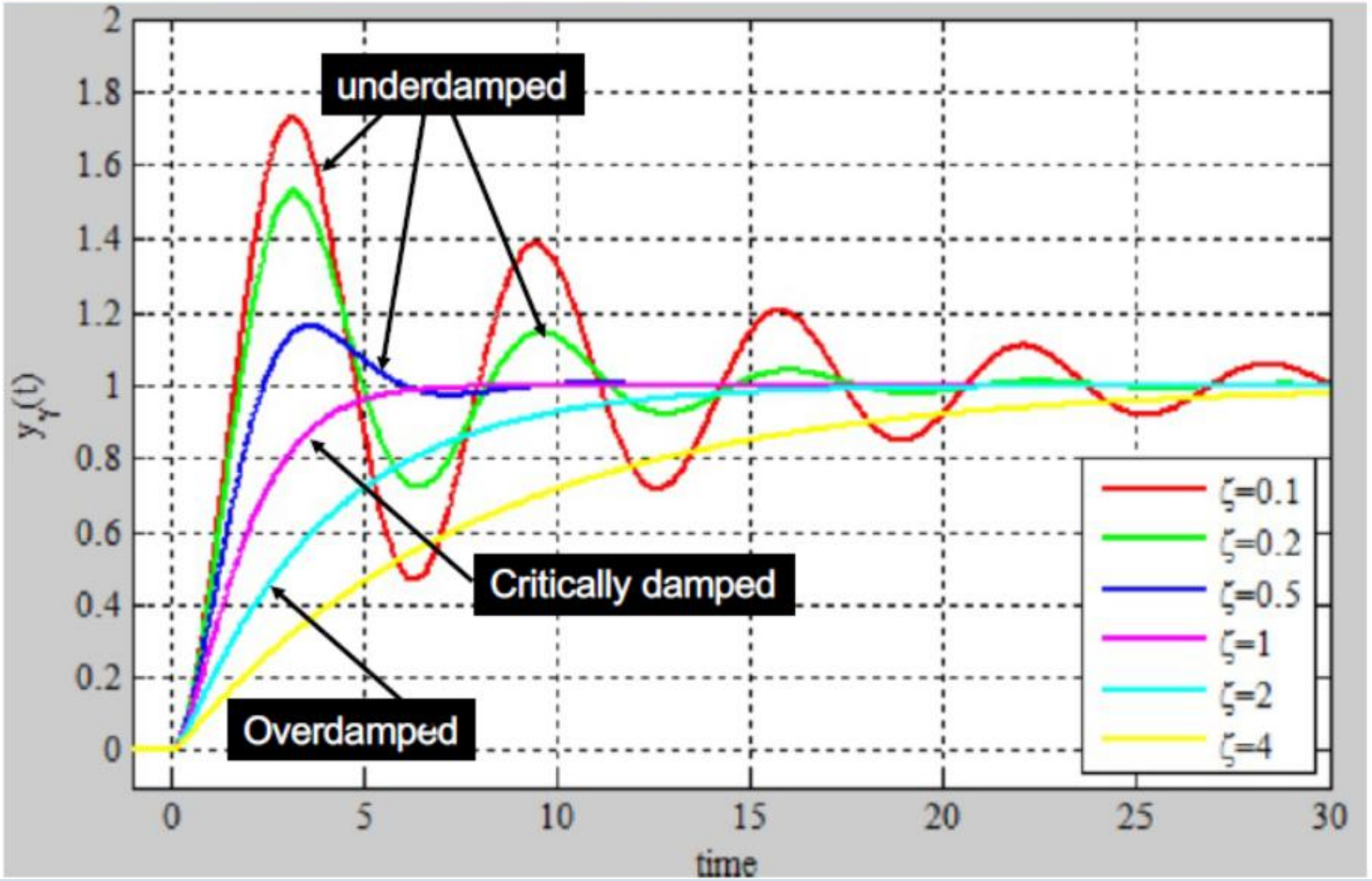
- Depending on the value of the damping factor ζ, there are five cases of interest, each having a specific behaviour:

$$H(s) = \frac{b_0}{s^2 + a_1 s + a_0} = K \frac{{\omega_0}^2}{s^2 + 2\zeta\omega_0 s + {\omega_0}^2}$$

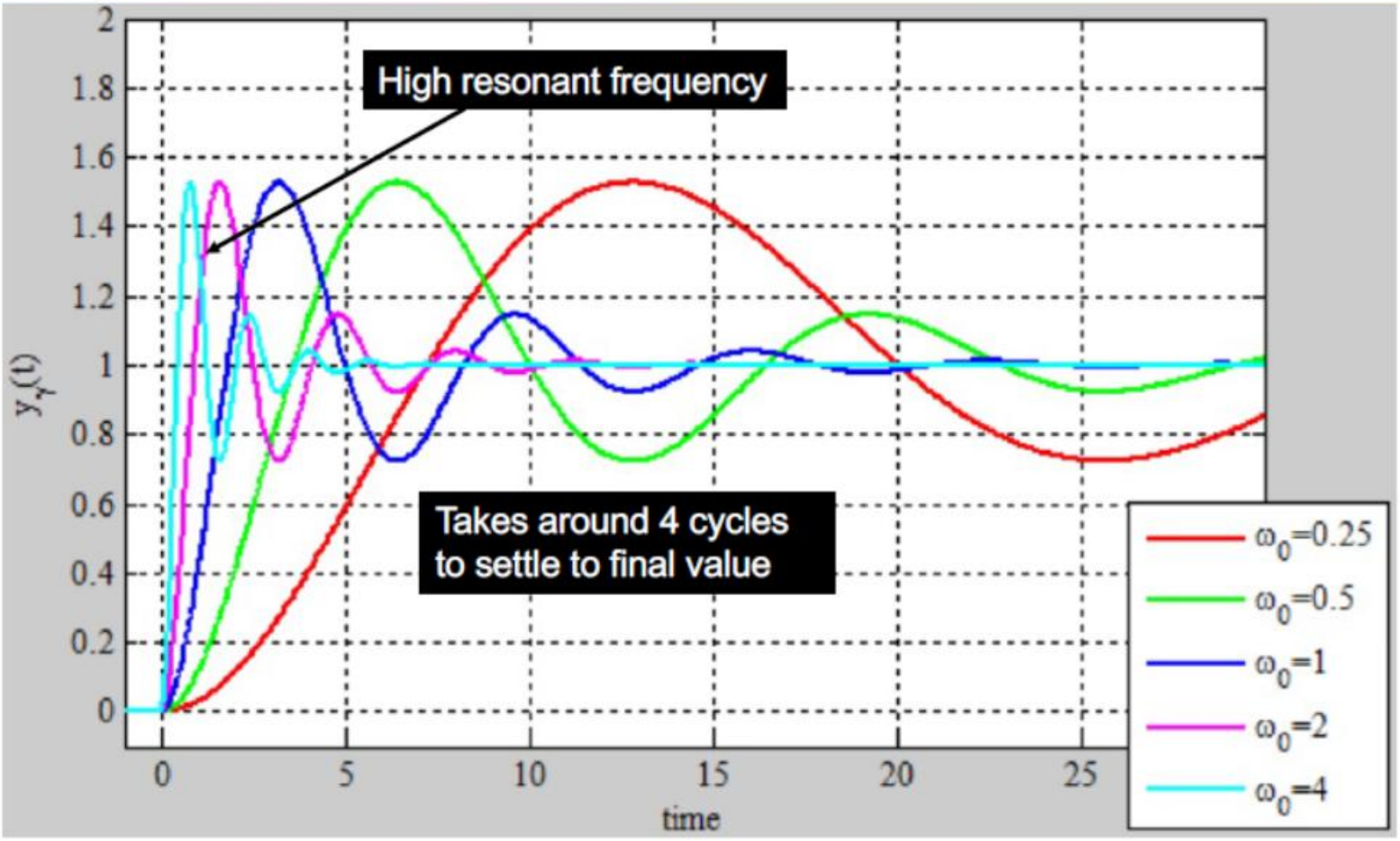
- Root of denominator: $s = -\zeta\omega_0 \pm \omega_0\sqrt{\zeta^2 - 1}$

Name	Value of ζ	Roots of s	Characteristics of "s"
Overdamped	ζ>1	$s = -\zeta\omega_0 \pm \omega_0\sqrt{\zeta^2 - 1}$	Two real and negative roots
Critically Damped	ζ=1	$s = -\omega_0$	A single negative roots
Underdamped	0<ζ<1	$s = -\zeta\omega_0 \pm j\omega_0\sqrt{1 - \zeta^2}$	Complex conjugate ($j = \sqrt{-1}$);
Undamped	ζ=0	$s = \pm j\omega_0$	Pure imaginary (no real part)
Exponential Growth	ζ<0	$s = -\zeta\omega_0 \pm \omega_0\sqrt{\zeta^2 - 1}$	Roots may be complex or real, but the real part of s is always positive

Step Response for different damping factors



Step Response at ω_0, $\varsigma = 0.2$



General Values

$$t_r \approx \frac{1.8}{\omega_n}$$

$$t_s \approx \frac{4.6}{\xi\omega_n} = \frac{4.6}{\sigma}$$

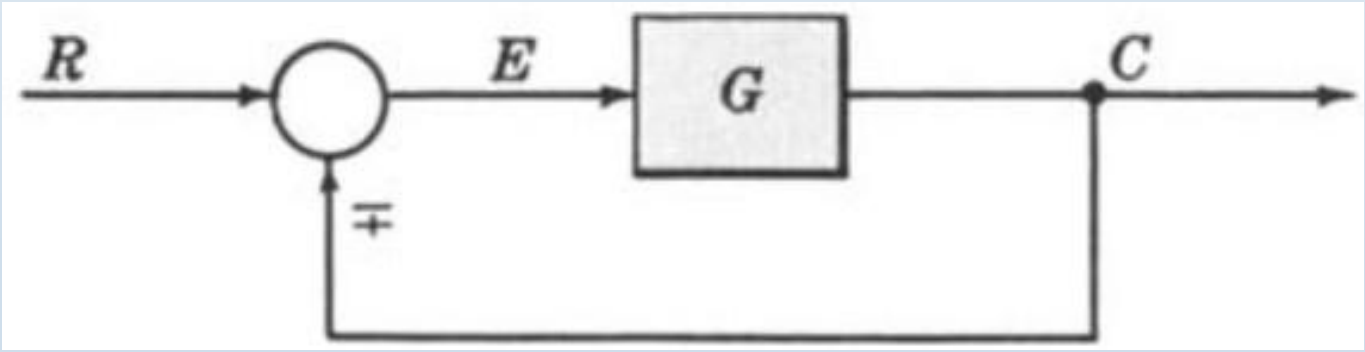
$$M_p \approx e^{-\pi\xi/\sqrt{1-\xi^2}} \quad (0 \le \xi \le 1) \qquad M_p \approx \left(1 - \frac{\xi}{0.6}\right) \quad (0 \le \xi \le 0.6)$$

For the overshoot M_p, two often used values are $M_p = 16\%$ for $\xi = 0.5$ and $M_p = 5\%$ with $\xi = 0.7$.

Quiz # 1

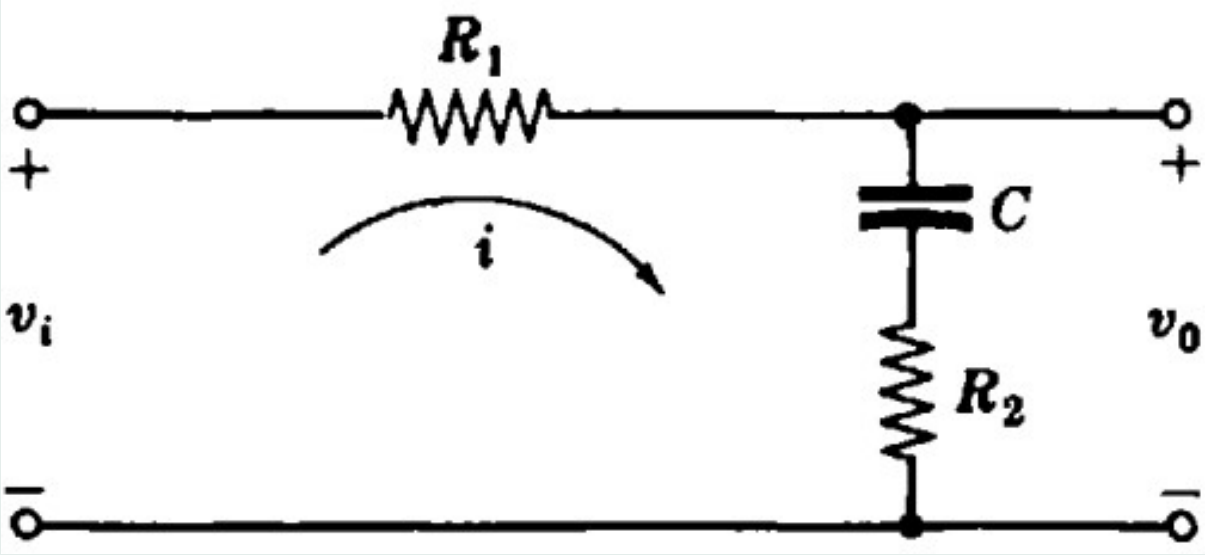
Determine C/R of the following Feedback Loop system

$C/R \equiv$ closed-loop transfer function $\equiv$ control ratio



Quiz # 2

1. Determine the transfer function of the R-C network?
2. Poles & Zero?
3. Draw Poles and Zeros?



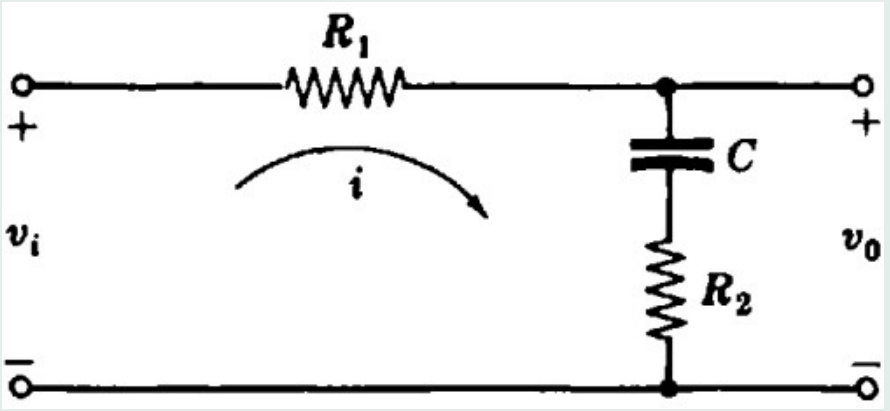
Quiz 2: Solution

$$iR_1 + \frac{1}{C}\int_0^t i\,dt + iR_2 = v_i$$

$$\left(R_1 + R_2 + \frac{1}{Cs}\right) I(s) = V_i(s)$$

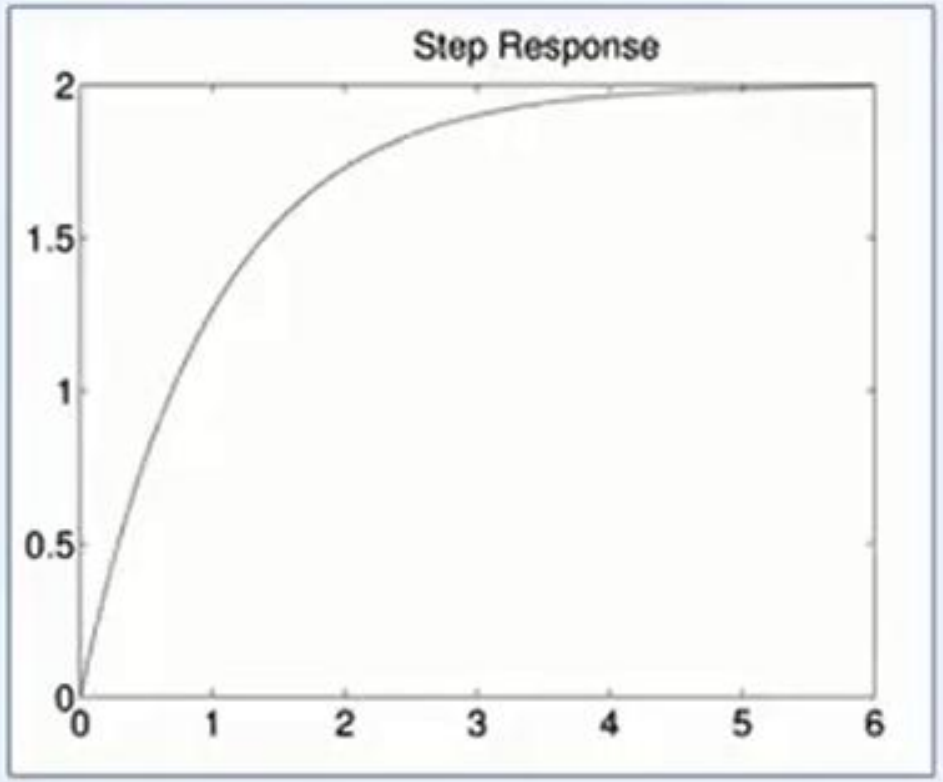
$$V_0(s) = \left(R_2 + \frac{1}{Cs}\right) I(s)$$

$$\frac{V_0(s)}{V_i(s)} = \frac{R_2 + 1/Cs}{R_1 + R_2 + 1/Cs}$$



Quiz # 3

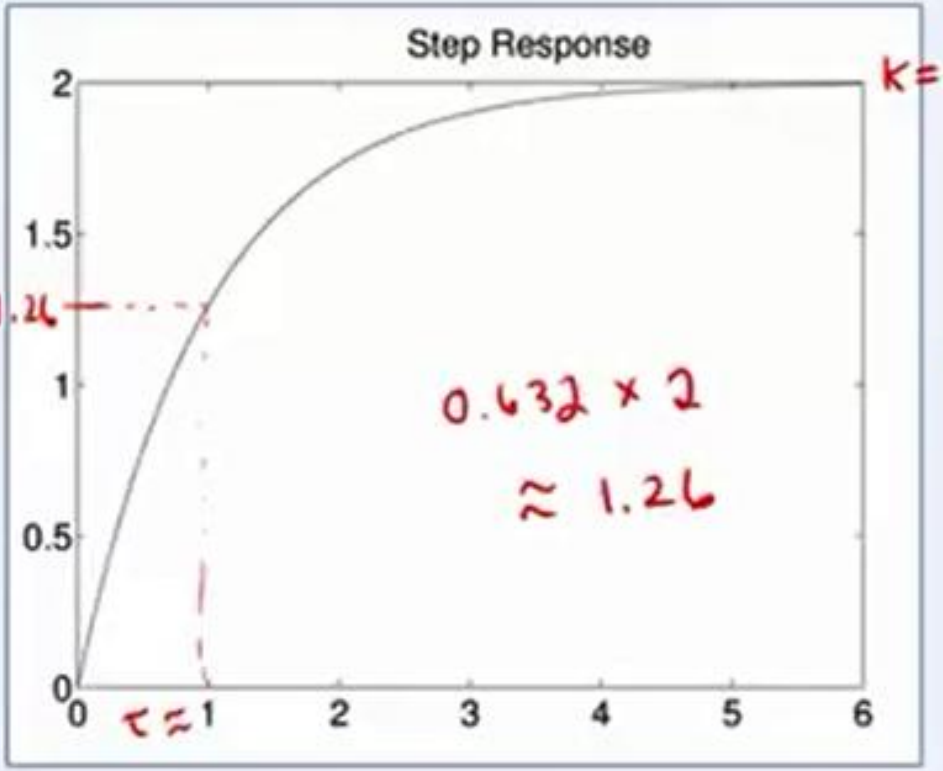
- Determine the TF of the system that produced the following output in response to a unit step input



Example

match form $\frac{K}{\tau s + 1}$

- Determine the TF of the system that produced the following output in response to a unit step input



$K = 2$ (DC gain)

$0.632 \times 2 \approx 1.26$

By inspection
$K \approx 2$
$\tau \approx 1$

$$\frac{Y(s)}{U(s)} = \frac{2}{s+1}$$

$$= \frac{1}{0.5s + 0.5}$$

First-Order Systems

- Basic form: $\tau \dot{y}(t) + y(t) = ku(t)$

$$\Rightarrow G(s) = \frac{Y(s)}{U(s)} = \boxed{\frac{k}{\tau s + 1}}$$

- Key parameters:

τ = time constant

k = DC gain

First-Order Systems

$$\frac{Y(s)}{U(s)} = \frac{K}{\tau s + 1} = G(s)$$

- Step response ($u(t) = 1(t)$) $\Rightarrow U(s) = \frac{1}{s}$

find $y(t)$

$$Y(s) = G(s)U(s) = \frac{K}{\tau s+1} \cdot \frac{1}{s} = \frac{A}{s} + \frac{B}{\tau s+1} = \frac{A(\tau s+1) + Bs}{s(\tau s+1)}$$

matching numerators $\Rightarrow$

s^1: $0 = A\tau + B \Rightarrow B = -A\tau = -K\tau$

s^0: $K = A$

$$y(t) = \mathcal{L}^{-1}[Y(s)] = \mathcal{L}^{-1}\left[\frac{K}{s} - \frac{K\tau}{\tau s+1}\right] = \boxed{K - Ke^{-t/\tau}, \; t \geq 0}$$

$\frac{K}{s+1/\tau}$

x_p x_h

First-Order Systems

$$y(t) = k(1 - e^{-t/\tau}), t \geq 0$$

For $t=\tau$

$$y(t) = k(1 - e^{-1}) = 0.632k$$

For $t=4\tau$

$$y(t) = k(1 - e^{-4}) = 0.98k$$

within 2% of final value is generally considered ss

